

INDU 211/2X — Assignment #1 Solutions

Course: INDU 211/2X — Introduction to Production & Manufacturing Systems

Instructor: Prof. Masoumeh Kazemi

Student: Mohamed Sharafath Shahul Hameed

Question #2 Selection: Option (a) — Assembling an IKEA Wooden Chair

Submission Requirement: Single PDF Document on Moodle

Question #1: Bill of Materials (BOM) — Bookcase (Fig. P3.2)

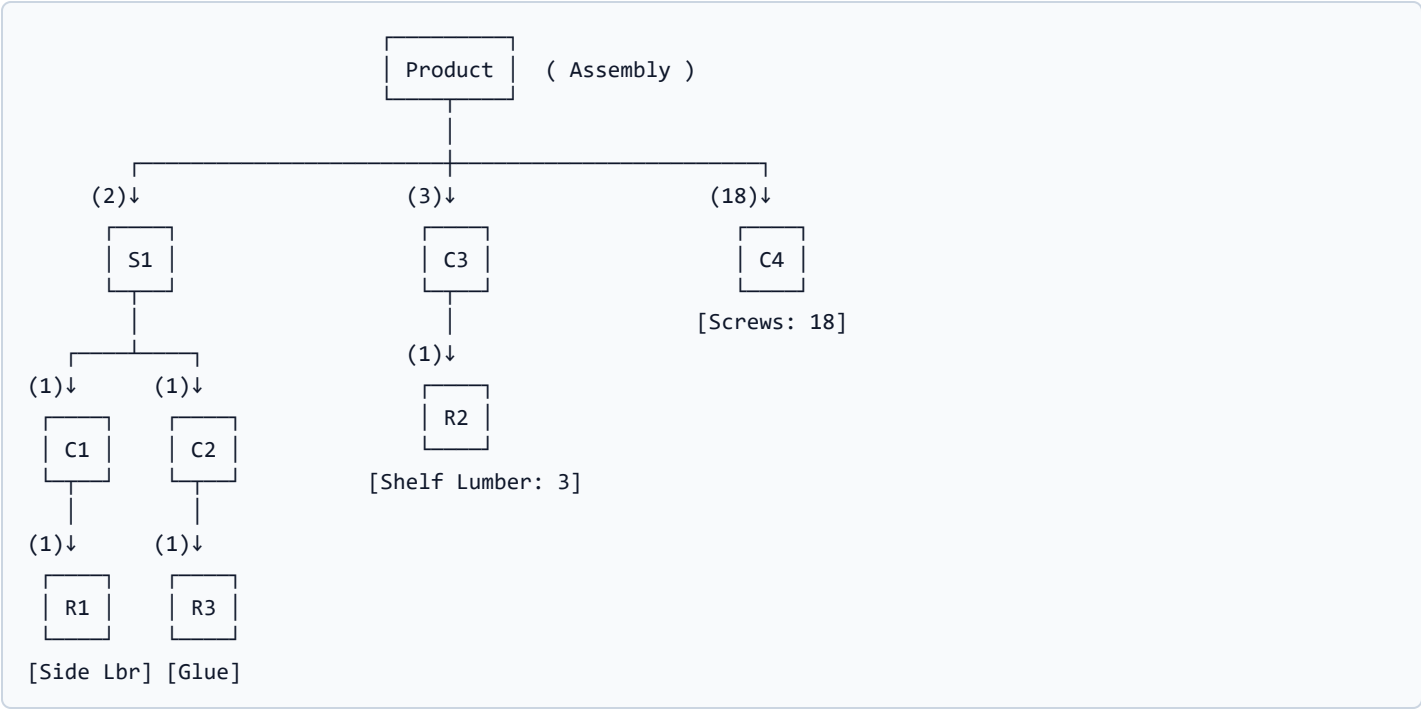
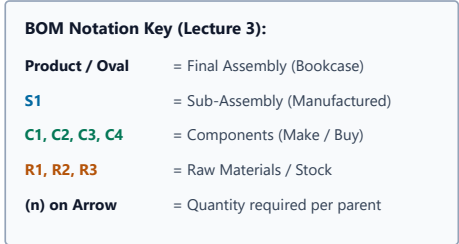
1.1 Product Description & Specifications

From **Figure P3.2 (Chapter 3, Problem 2)**, the product is a manufactured **wooden bookcase** with the following technical specifications: - **Sides (2):** Wooden panels measuring $3'2'' \times 2' \times \frac{3}{4}''$ ($38'' \times 24'' \times 0.75''$). - **Shelves (3):** Wooden panels measuring $2' \times 2' \times \frac{3}{4}''$ ($24'' \times 24'' \times 0.75''$). - **Joints:** All shelf-to-side joints are dadoed to a depth of $\frac{3}{8}''$ ($0.375''$) on the inner face of the sides. - **Fasteners:** Screws (3 per side per shelf) = $3 \times 2 \times 3 = \mathbf{18}$ screws total (purchased standard hardware). - **Decals:** (2) University Decals glued onto the exterior of the two side panels.

1.2 Graphical BOM Product Structure Tree (Lecture 3 Standard)

The Bill-of-Materials product structure tree is formatted according to the exact convention in **Lecture 3 on Manufacturing Engineering (Slide 7)**: - **Product / Oval:** Final assembled product (Assembly) - **S1 :** Sub-assembly (Side Panel Sub-Assembly) - **C1 , C2 , C3 , C4 :** Components (Make or Buy) - **R1 , R2 , R3 :** Raw materials / stock items - **(n) on Arrows:** Quantity required per parent assembly

Product Structure Tree for Bookcase (Fig. P3.2) — INDU 211 / Chapter 3 Standard



1.3 Master Bill of Materials (BOM) Table

Level	Code	Item Description	Qty / Parent	Total Qty	Make / Buy	Material / Dimensions / Processing
0	Product	Bookcase Assembly	—	1	Make	Final assembly of side sub-assemblies, shelves, and screws
1	S1	Side Panel Sub-Assembly	2	2	Make	Left & Right side sub-assemblies (Panel + Decal glued)
2	C1	L Side Panel (Wood)	1	2	Make	Lumber $3'2" \times 2' \times \frac{3}{4}"$, 3 dado grooves cut $\frac{3}{8}"$ deep, 9 screw holes pre-drilled
3	R1	L Lumber Board	1	2	Buy	Raw board stock $3'2" \times 2' \times \frac{3}{4}"$ (\$12.67\text{ sq ft}\$ total)
2	C2	L University Decal	1	2	Buy	Standard adhesive university graphic/emblem
3	R3	L Adhesive / Glue	1	2	Buy	Wood / laminate bonding adhesive
1	C3	Shelf	3	3	Make	Lumber $2' \times 2' \times \frac{3}{4}"$, cut to length with square ends
3	R2	L Lumber Board	1	3	Buy	Raw board stock $2' \times 2' \times \frac{3}{4}"$ (\$12.00\text{ sq ft}\$ total)
1	C4	Screws (Wood Fasteners)	18	18	Buy	Standard flat-head wood screws ($3 \times \text{shelf} \times \text{side} = 3 \times 2 \times 3 = 18$)

1.4 Raw Material Summary & Calculations

Raw Material Item	Code	Unit Dimensions	Calculation	Total Quantity Required
Lumber for Sides	R1	$3.167' \times 2' \times 0.75"$	$2 \times \text{panels} \times (3.167' \times 2') = 12.67 \text{ sq ft}$	\$12.67\text{ sq ft}\$ (\$~13\text{ sq ft}\$ gross)
Lumber for Shelves	R2	$2' \times 2' \times 0.75"$	$3 \times \text{shelves} \times (2' \times 2') = 12.00 \text{ sq ft}$	\$12.00\text{ sq ft}\$
Total $\frac{3}{4}"$ Lumber	—	$\frac{3}{4}"$ board thickness	$12.67 + 12.00 = 24.67 \text{ sq ft}$	\$~25\text{ sq ft}\$
Wood Screws	C4	Standard wood screws	$3 \times \text{shelf} \times \text{side} \times 3 \times 2 = 18$	18 screws
University Decals	C2	Glued emblem	$1 \times \text{side} \times 2 \times \text{sides} = 2$	2 decals
Wood Glue / Adhesive	R3	Commercial adhesive	As required for decal application	Consumable

Question #2: Assembly of an IKEA Wooden Chair (Option a)

2.1 — Product Description & Key Assembly Activities

Product Selected: IKEA STEFAN Wooden Dining Chair (flat-pack furniture requiring customer/assembly-line production).

Main Components Catalog

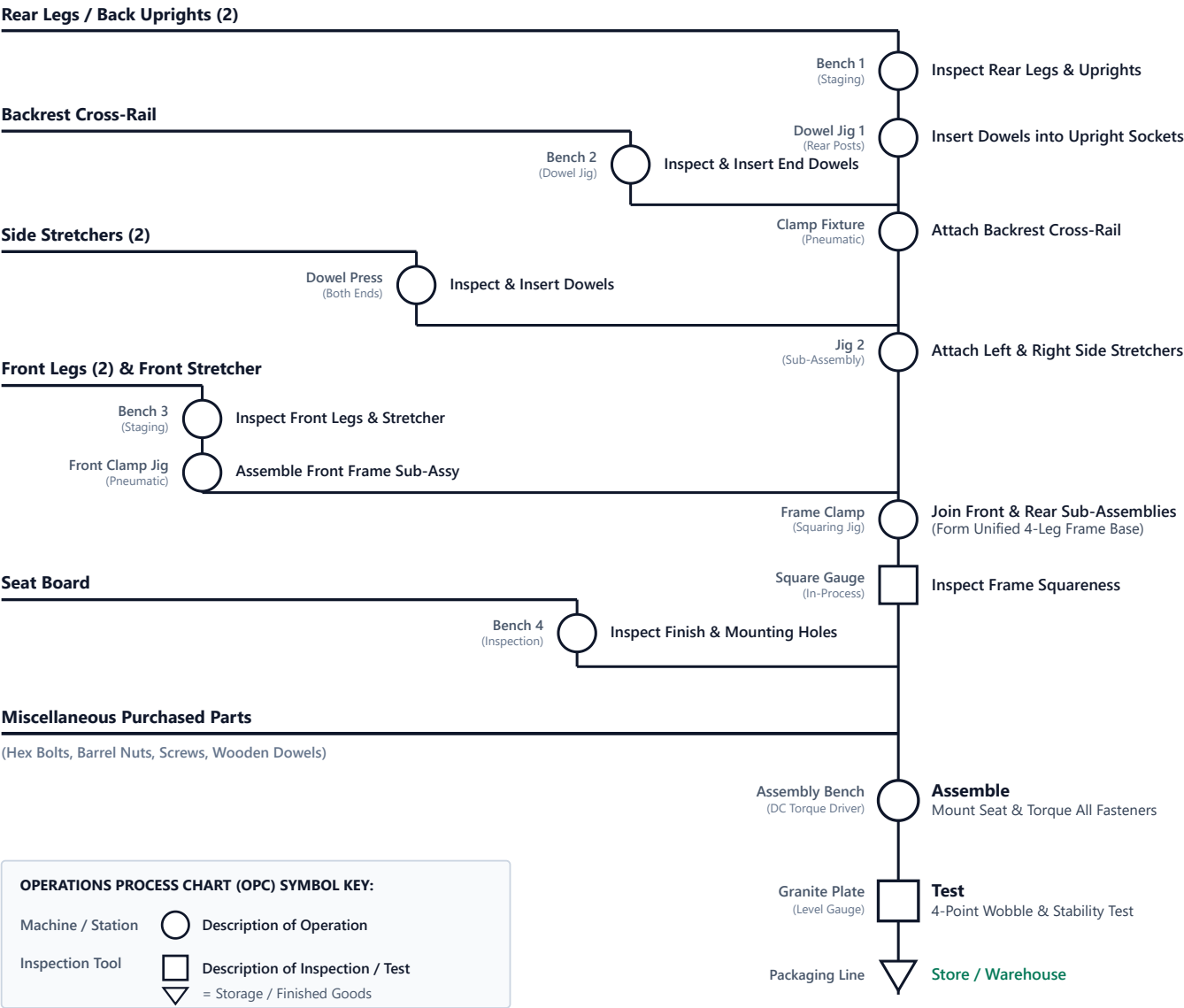
#	Component	Qty	Function & Technical Details
1	Seat Board	1	Pre-finished solid pine seat board with pre-drilled counterbored mounting holes
2	Front Legs	2	Turned front wooden legs with pre-drilled dowel sockets
3	Rear Legs / Back Uprights	2	Full-height uprights forming both the rear legs and backrest support frame
4	Backrest Cross-Rail	1	Curved ergonomic horizontal lumbar cross-rail connecting rear uprights
5	Side Stretchers	2	Horizontal braces connecting front legs to rear uprights for lateral rigidity
6	Front Stretcher	1	Horizontal brace connecting both front legs
7	Wooden Dowels	12	Fluted wooden pins for joint alignment and shear strength
8	Hex Bolts & Barrel Nuts	4–6	Steel fasteners securing the seat board to the perimeter frame
9	Allen Key (Hex Wrench)	1	Assembly tool provided in package for torque application

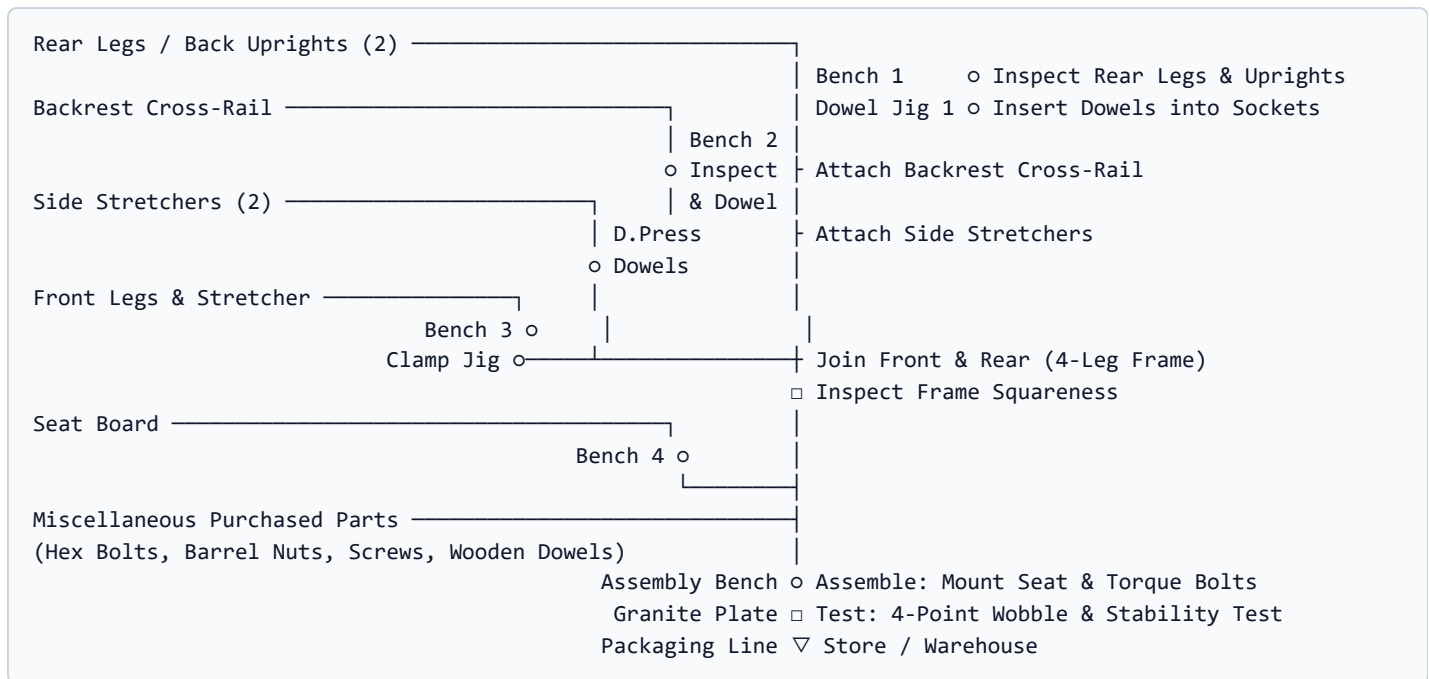
2.2 — Operations Process Chart (OPC) — Lecture 3 Standard

The Operations Process Chart below follows the exact industrial engineering convention demonstrated in **INDU 211 Lecture 3 (Slide 20, "Operations process chart for check valve")**: - **Main Assembly Trunk (Far Right)**: The principal structural member (**Rear Legs / Back Uprights**) forms the vertical backbone on the far right. - **Sub-Assembly Feeder Lines (Left)**: Components enter horizontally from the left and drop down through their respective preparatory operations before merging into the main trunk. - **Miscellaneous Purchased Parts**: Hardware items (bolts, barrel nuts, screws, dowels) enter horizontally across all lines directly prior to final assembly. - **Symbol Standard**: - Circles (\$\bigcirc\$) represent **Operations** (Machine / Station on left, Description on right). - Squares (\$\square\$) represent **Inspections / Tests** (Inspection tool on left, Description on right). - Inverted Triangle (\$\nabla\$) represents **Storage / Finished Goods**.

▪ Operations process chart for wooden chair

IKEA STEFAN Wooden Dining Chair — INDU 211 Operations Process Chart (OPC) Standard





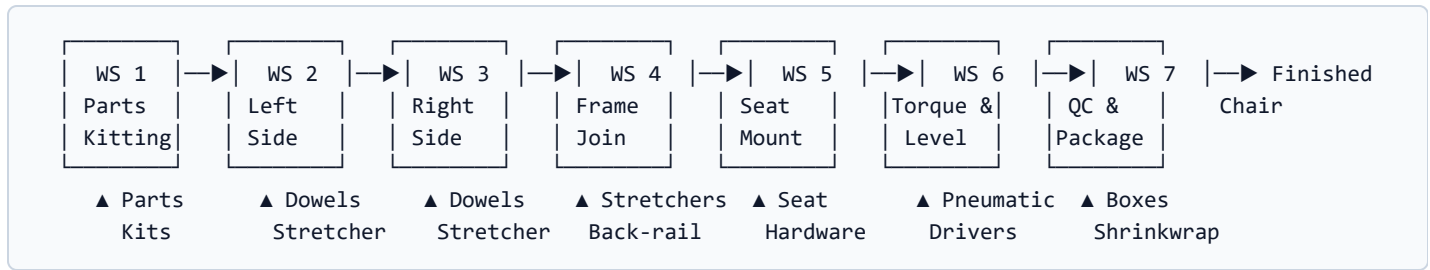
Operations Process Summary Table

Step #	Station / Tool	Symbol	Operation / Inspection Description	Output Sub-Assembly
1	Bench 1 (Staging)	\$\squarecirc\$	Inspect Rear Legs & Upright Sockets	Verified rear posts
2	Dowel Jig 1	\$\squarecirc\$	Insert Dowels into Upright Mortises	Pre-doweled rear posts
3	Bench 2 (Dowel Jig)	\$\squarecirc\$	Inspect & Insert End Dowels into Back Rail	Ready backrest rail
4	Clamp Fixture (Pneumatic)	\$\squarecirc\$	Attach Backrest Cross-Rail to Rear Uprights	Upper backrest sub-assy
5	Dowel Press (Dual Head)	\$\squarecirc\$	Inspect & Insert Dowels into Both Stretcher Ends	Ready side stretchers
6	Jig 2 (Sub-Assembly)	\$\squarecirc\$	Attach Left & Right Side Stretchers to Rear Posts	3-sided rear structure
7	Bench 3 (Staging)	\$\squarecirc\$	Inspect Front Legs & Front Stretcher	Staged front parts
8	Front Clamp Jig	\$\squarecirc\$	Assemble Front Legs with Front Stretcher	Front frame sub-assy
9	Frame Clamp (Squaring Jig)	\$\squarecirc\$	Join Front Sub-Assy to Rear Structure	Unified 4-leg base frame
10	Square Gauge	\$\square\$	In-Process Inspection of Frame Squareness & Joint Flushness	Quality-checked frame
11	Bench 4 (Visual QC)	\$\squarecirc\$	Inspect Seat Board Finish & Align Bolt Holes	Staged seat board
12	Feeder Line	—	Feed Miscellaneous Purchased Hardware (Bolts, Nuts, Dowels)	Hardware kit staged
13	Assembly Bench (DC Driver)	\$\squarecirc\$	Assemble: Lower seat onto frame, insert bolts into barrel nuts, torque to spec	Fully assembled chair
14	Granite Surface Plate	\$\square\$	Test: 4-point flatness, stability, and load wobble test	Certified product
15	Packaging Conveyor	\$\squaretriangledown\$	Protective wrap, pack, and transport to warehouse	Finished goods inventory

2.3 — Assembly Line Design & Economic Factors (Chapter 3 Standard)

2.3a — Assembly Line Arrangement (Product Layout)

To mass-produce IKEA-style chairs, a **continuous progressive assembly line** (linear product layout) is arranged with 7 dedicated workstations in series:



Station	Task Description	Tooling / Fixture	Standard Time
WS 1	Parts Kitting: Unpack, sort and stage components into mobile fixtures	Gravity bins	30 sec
WS 2	Left Side Assembly: Insert dowels, join left stretcher to legs	Pneumatic press clamp	45 sec
WS 3	Right Side Assembly: Mirror of WS 2 for right frame	Pneumatic press clamp	45 sec
WS 4	Frame Consolidation: Join front & rear frames with stretchers	Squaring clamp jig	50 sec (Bottleneck)
WS 5	Seat Mounting: Lower seat board, align holes, insert barrel nuts	Alignment pins	40 sec
WS 6	Fastener Torquing: Torque all bolts with automatic shut-off driver	Overhead DC driver	35 sec
WS 7	Inspection & Packaging: 4-point wobble test, pack into box	Granite surface plate	35 sec

- **Line Balancing & Cycle Time:** The line bottleneck is WS 4 (50 seconds/unit), setting the production cycle time to $CT = 50 \text{ seconds/unit}$.
- **Daily Throughput (Capacity):** On a standard 8-hour shift (28,800 seconds), daily production capacity is:

$$\text{Daily Output} = \frac{28,800 \text{ seconds}}{50 \text{ seconds/unit}} = 576 \text{ chairs/day}$$

2.3b — Economic Factors to Consider (Cost-Volume & Break-Even Analysis)

Based on the **Cost-Volume Relationships** taught in **INDU 211 (Chapter 3, Slides 10–17)**, the total cost of setting up and operating an assembly line follows the linear cost function:

$$Y = a \cdot X + b \quad \Longleftrightarrow \quad TC = VC \cdot Q + FC$$

Where: - Y (or TC) = Total Production Cost (\$) - X (or Q) = Production Volume (number of chairs) - a (or VC) = Variable Cost per chair (\$/unit) - b (or FC) = Fixed Cost (\$)

The **4 essential economic factors** to evaluate for this assembly line are:

1. Fixed Costs (\$FC\$ or \$b\$) — Capital Investment

These are one-time capital expenditures that must be paid **regardless of how many chairs are produced**: - **Conveyor system & workstations:** Purchasing and installing the roller conveyor line. - **Dedicated jigs & pneumatic clamps:** Custom fixtures at WS 2, WS 3, WS 4 to hold chair legs square during joining. - **Tooling:** Overhead pneumatic screwdrivers and torque drivers. - **Facility space:** Factory floor space lease, utility setup, and lighting. - *Estimated Fixed Cost for chair line:*
 $FC = \$150,000$

2. Variable Costs (\$VC\$ or \$a\$) — Per-Unit Recurring Costs

These costs increase directly with every chair assembled: - **Raw materials & components:** Flat-pack chair lumber parts, wooden dowels, hex bolts, barrel nuts ($\sim \$14/\text{chair}$). - **Direct labor:** Operator assembly wages distributed per unit ($\sim \$5/\text{chair}$). - **Utilities & tool wear:** Electricity for pneumatic compressors, bit replacement ($\sim \$1/\text{chair}$). - **Total Variable Cost per unit:** $\mathbf{VC} = \$20/\text{chair}$.

3. Break-Even Point (\$BEP\$) — Decision Threshold Rule

As established in Lecture 3 (Slide 12), the Break-Even Point determines the minimum sales volume required to recover fixed setup costs:

$$\mathbf{Q_{BEP}} = \frac{FC}{P - VC} = \frac{b}{P - a}$$

Assuming a wholesale selling price of $\mathbf{P = \$50/\text{chair}}$: $\mathbf{Q_{BEP}} = \frac{\$150,000}{\$50 - \$20} = \frac{\$150,000}{\$30} = \mathbf{5,000 \text{ chairs}}$

Core Decision Rule (INDU 211 Slide 12):

- If **projected market demand is greater than 5,000 chairs ($Q > Q_{BEP}$)**: Setting up the dedicated assembly line is economically justified and profitable.
- If **projected demand is less than 5,000 chairs ($Q < Q_{BEP}$)**: The company should **not** invest in the assembly line, because the high fixed costs cannot be recovered.

4. Process Selection & Economies of Scale (Manual Bench vs. Assembly Line)

As shown in INDU 211 Slide 13–15 (Process Comparison): - **Manual Batch Assembly (Job Shop)**: Has very low fixed costs ($FC_1 = \$10,000$), but high variable labor costs ($VC_1 = \$35/\text{chair}$) because one worker builds each chair manually. - **Continuous Assembly Line**: Has high fixed costs ($FC_2 = \$150,000$), but low variable costs ($VC_2 = \$20/\text{chair}$) due to high specialization and speed. - **Crossover Volume**: $\$10,000 + 35 \cdot Q = 150,000 + 20 \cdot Q \implies 15 \cdot Q = 140,000 \implies \mathbf{Q = 9,333 \text{ chairs}}$ - **Conclusion**: For small batches ($Q < 9,333$), manual assembly is cheaper. For mass production ($Q > 9,333$), the dedicated assembly line produces chairs at the lowest total cost per unit due to **Economies of Scale**.

End of Assignment #1 Solution — Prepared for INDU 211/2X Submission